

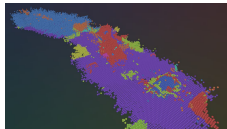
Pixelwise Image Feature Extraction

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Motivation

- Open-vocabulary maps with a queryable, dense spatial representation are desirable for smart, high-level semantic planning
- Use case of drone navigation in unknown environments

Figure 1 - PCA visualization of feature-vectors assigned to map voxels. Initial iteration of project outcomes.
Credit: Omar Alma (DSTA, AirLab)



Background

- Current image-language alignment models like CLIP encode text and images in the same latent space but are image-wise rather than pixel-wise

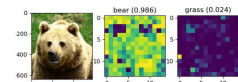


Figure 2 - Poor locality of CLIP image features across output patches

- Without post-processing, SOTA methods like NACLIP and PixelCLIP produce noisy results

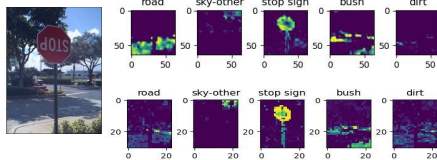


Figure 3 - Pixelwise cosine similarities with CLIP-encodings of text prompts using features generated by PixelCLIP (top) and NACLIP (bottom)



Current Strategy

- Take image crops using SAM2 segmentation masks
- Use CLIP image encoder to extract feature vector for each crop
- Interpolate feature vectors over masked pixels in original image



Figure 4 - Original image (left), image crops (step 1, center), feature vectors (step 3, right)

CLIP Input Parameters		SAM2 Segmentation Parameters		Results	
Crop Background	Broadcast Output Type	Box NMS Thresh	Pred IOU Thresh	mIoU	
Average	Mask	0.7	0.5	21.04%	
Average	Mask	0.7	0.7	19.44%	
Average	Mask	0.25	0.7	18.41%	
Black	Mask	0.7	0.7	16.59%	
Crop	Mask	0.7	0.7	13.53%	
Average	Crop	0.7	0.7	12.28%	

Table 1 - Ablation using mean intersection over union (mIoU) of open-vocabulary label queries over our system's feature vectors relative to ground-truth segmentation labels

Results

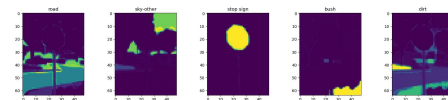
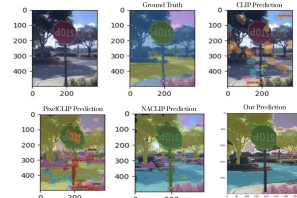


Figure 5 - Pixelwise cosine similarities with CLIP-encodings of text prompts using features generated by our system

Figure 6 -

Comparison of maximum pixel feature similarities to set of encoded text labels using different systems' pixel labels



Model	mIoU (%)
PixelCLIP	20.2
NACLIP	20.5
CLIP	3.8
SAM2-based (Ours)	21.14

Table 2 - mIoU comparison of our pixel-labelling method compared to two existing SOTA methods without post-processing on COCO-Stuff 2017 validation dataset

Future Direction

Improve Teacher network

Areas to explore: Masking background pixels directly in CLIP attention layers, pooling results over multiple encoders, agglomerative clustering

Train Student network

CLIP Input Parameters		SAM2 Segmentation Parameters				System Parameters	Results	
Input Background	Broadcast Output Type	Box NMS Thresh	Stability Score Thresh	Stability Score Offset	Pred IOU Thresh	Interpolation Size	mIoU	Time (s/frame)
Average	Masked	0.7	0.75	0.5	0.5	224	21.04%	0.875
Average	Masked	0.7	0.75	0.7	0.7	64	19.44%	0.768
Average	Masked	0.25	0.75	0.7	0.7	224	18.41%	0.746
Average	Masked	0.7	0.95	0.7	0.7	224	17.77%	0.659
Black	Masked	0.7	0.75	0.5	0.7	224	16.59%	1.31
Crop	Masked	0.7	0.75	0.5	0.7	224	13.53%	3.523
Average	Cropped	0.7	0.75	0.5	0.7	224	12.28%	0.813

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